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INTRODUCTION

By organizing resources, scheduling, procurement, and execution throughout several planning periods, PPC is a strategic talent that transforms erratic demand into realistic commitments (Slack et al., 2019). It is particularly crucial because construction is project-based, delivery is contractual, and performance is affected by both external and internal factors, including weather and supply shortages, as well as internal factors, like different subcontractors and temporary setups (Kerzner, 2022; Koskela, 2000). Poor PPC not only lowers efficiency but also causes rework, delays, safety concerns, and decreased profits as a result of rushed work and disruptions (Stevenson, 2021). This research examines NordBuild Construction GmbH (NBC), a German contractor that works on municipal, commercial, and residential projects, from the perspective of a consultant. Uncertain tender pipelines, shifting customer needs, a lack of experienced workers, fluctuating lead times and material costs, and regulatory issues are all problems for NBC.

CHAPTER ONE

1. PPC's strategic importance in NBC

1.1 Why PPC is strategic (not administrative) in construction

In the case of NBC, PPC is tactical in the sense that its competitiveness is pegged on plausible promises and perceived risk. The value of construction is created by distinctive projects having definite milestones and contractual implications (e.g. liquidated damages, performance deductions) that turns reliability and change-handling into the primary focus of profitability (Kerzner, 2022). PPC is thus a risk form of governance: it determines whether or not NBC is able to take on new work without over-booking essential trades, whether or not they are able to absorb disruptions without breaking down productivity, and whether or not they are able to ensure they maintain the margin constrained by disciplined procurement and resource allocation (Slack, Brandon-Jones and Johnston, 2019). The over-deterministic plans fail to consider the fact that site activities are very variable and interdependent; variability spreads across handovers and precludes downstream activities (Koskela, 2000).

1.2 PPC functions mapped to NBC's real operating system

The core is normally found in PPC which is framed through planning, routing, scheduling, dispatching and control.

These functions in NBC are:

Planning (portfolio + project): transform the tender pipeline and successful projects into a workload plan, allocated labour hours, subcontractor package, and priorities on procurement. This is equivalent to aggregate planning and S&OP logic, but project portfolios (Slack, Brandon-Jones and Johnston, 2019). Routing (method + work package): select construction methods and work packages (e.g. cast-in-place/ precast; facade system; phasing plan). The choice of methods alters cycle time, lead times, risk profile and resource requirements- so it should be acted upon as PPC lever rather than a decision based on technical considerations (Koskela, 2000).

Scheduling (baseline + rolling lookahead): establish a CPM baseline in which the contractual control is to be exercised, followed by running a rolling lookahead planning to make the work ready and avoid variability in execution (Kerzner, 2022; Ballard, 2000).

Release of tasks (task release): release weekly/daily tasks when they have prerequisites (design information, permits, access, materials, equipment, predecessor completion). This bridge is the operational plan-workface productivity (Ballard, 2000).

Follow-up and control: measure progress and productivity variance, refreeze forecasts and constraints, and activate replanning. No feedback makes the plan a fiction; feedback makes PPC a learning system (Stevenson, 2021).

Communication is important: When portfolio planning overallocates a fixed trade (e.g., MEP installers), the schedule of the project will be systematically infeasible; dispatching will turn into firefighting, which will raise waiting times and cut productivity; that aggravates the delay risk and leads to expediting and rework, which damage profitability (Slack, Brandon-Jones and Johnston, 2019; Kerzner, 2022).

1.3 NBC's sector-specific challenges and the PPC design implications

The PPC of NBC will have to be influenced by construction-specific limitations: Tender pipeline volatility (demand is probabilistic)

1. Demand begins as bids. The fact that bids are treated as firm demand and only signed contracts are treated as demand leads to overbooking and underutilisation respectively. PPC should consequently apply probability-weighted forecasting on a portfolio level, and time fences after having won a project (Stevenson, 2021).

2. Fragmented supply chain and reliance of subcontractor. Capacity is partly external. PPC will have to incorporate subcontractor availability and reliability into the plan, otherwise schedules will be aspirational. At the operation level, NBC is organizing an interconnected system of production, in which the alignment and lead-time management is strategic (Slack, Brandon-Jones and Johnston, 2019).

3. Trades based on bottlenecks and common resources. There are some trades that serve as bottlenecks such as crane time, form work crews, and MEP coordination. To avoid systemic delays, which are reflected by the logic that performance of a system is constrained, rather than capacity, PPC should focus on and safeguard these constraints across projects (Hopp and Spearman, 2011).

4. Variability and rework risk Variability caused by changes in design, inspection delays and weather interruptions flows through dependencies. NBC must have rolling constraint elimination, buffer policy and dedicated task preparation to stabilize flow and lessen making-do (Ballard, 2000; Koskela, 2000).

5. Sustainability/compliance constraints HSE laws, noise limits, waste segregation and carbon reporting affect viable working periods and processes; PPC needs to consider these limits at the start or risk impractical plans and reactive resequencing (Slack, Brandon-Jones and Johnston, 2019).

1.4 Strategic PPC positioning for NBC

NBC needs to operate PPC as a two-level system:

1. Portfolio control (S&OP-like): probability-weighted demand, master capacity plan of critical trades and equipment, subcontractor framework capacity and long-lead procurement risk planning.

2. Project control (execution): CPM base + rolling lookahead and constraint-based task release (Last Planner logic), which are facilitated by strict progress measurement and learning loops. This is in line with the purpose of strategic capability of PPC, which better delivery reliability and resilience and shield margin in case of uncertainty (Kerzner, 2022; Ballard, 2000).

CHAPTER TWO

2.Demand management, forecasting and "construction MPS"

2.1 Independent vs dependent demand in NBC (clear, practical)

In NBC's context: Clients/market provide independent demand: projects are won (sign contracts), change orders, and (where applicable) service/maintenance requests (Stevenson, 2021). Trade-based on BoQ/BOM, work packages and methods: concrete volume, rebar tonnage, facade units, labour hours by trade, crane days, subcontractor packages - time-phased by the schedule: dependent demand is derived with respect to independent demand (Slack, Brandon-Jones and Johnston, 2019). This is significant since NBC is expected to project independent demand (project starts/workload) followed by the explosion of dependent demand in terms of quantities and techniques in line with MRP logic that involves end demand as a drive in component requirements (Vollmann et al., 2005).

2.2 Forecasting method: hybrid portfolio forecast (fit-for-construction)

Even one statistical technique (e.g. moving average of revenue) can be rendered useless in construction due to historical output being biased towards project mix and not patterns of repeatable demand. NBC requires a hybrid method:

1. Medium-term (3-12 months): probability-weighted pipeline tender forecast. Project value x probability of win x planned workload curve = S (Project value x probability of win x planned workload curve). This puts managerial discipline to the fore and minimizes optimism bias in capacity planning (Stevenson, 2021).

2. Short-term (0-12 weeks): confirmed workload exponential smoothing. After the signing of projects, NBC applies exponential smoothing to stabilise near-term planning of labour/equipment but with updates being allowed as variances on progress become known (Stevenson, 2021).

2.3 How forecast feeds the "MPS" in construction: Master Project & Resource Schedule

There is no MPS in the factory sense in construction, the functional equivalent of this is a Master Project Schedule (MPrS) and a Master Resource Schedule (MRgS) of constrained shared resources. The reasoning is the same create a demand into a workable time-based plan within capacity constraints (Slack, Brandon-Jones and Johnston, 2019). A representative Master Resource Schedule addressing the major constraint trades and crane time at NBC (representative numbers, to be replaced subsequently with your numbers) is shown below:

Master Resource Schedule (monthly, portfolio level)

Month	Confirmed workload (€m)	Expected workload (pipeline €m)	Total forecast (€m)	Formwork crews available (crew-months)	MEP crews available	Crane availability (days)	Feasibility assessment
Mar	6.0	1.5	7.5	18	12	40	Tight (MEP risk)
Apr	5.5	2.0	7.5	18	12	40	Tight (crane peaks)
May	5.0	2.5	7.5	18	12	40	High risk (MEP + crane)

What this achieves (consultant logic):

It makes the forecast operational into a capacity conscious plan (RCCP logic). It points out limitations in advance (MEP and crane peaks), which makes it possible to make decisions in advance: subcontractor framework agreements, staggered starts, alternative techniques (prefab), or negotiated milestone buffers (Hopp and Spearman, 2011; Kerzner, 2022).

2.4 Time fences to control "schedule nervousness"

In order not to have to resequence all the time and kill productivity of sites, NBC ought to adopt time gates like those of MPS control:

Frozen zone (0-4 weeks): the only changes permitted are safety/critical one; the materials and subcontractors are locked up. Slushy zone (5-12 weeks): partial flexibility with consent; the smoothing of resources is also allowing.

Liquid zone (>12 weeks): flexible; this is where pipeline uncertainty is actually taken. Time fences, decrease planners nervousness and establish stability without rejection of change (Vollmann et al., 2005).

2.5 Trade-offs: sensitivity, stability and capacity (explicit, as required)

NBC needs to demonstrate that it realizes what trade-offs: Responsiveness vs stability Accepting late change of client Enhances customer responsiveness at the expense of resequencing, waiting, and flow loss; task readiness discipline is capable of reducing chaos whilst still permitting controlled change (Ballard, 2000).

Capacity utilisation vs reliability in delivery: stretching crews to close to 100% capacity utilisation decreases flexibility; buffers in constraint resources are small ones and can enable greater reliability as well as lower the total cost of delay/rework (Hopp and Spearman, 2011).

Accuracy of the forecast vs cost of governance: probability-weighted forecasting is more realistic but governance is needed (bid probabilities, stage gates). In the absence of disciplined inputs, forecasting is more of a political process, not an analytical one (Stevenson, 2021).

Advice to the consultant: NBC needs to make PPC performance results performance KPI output: milestone OTIF, lookahead plan reliability (PPC in the terms of Last Planner), constraint removal rate, and productivity variance to plan, not as a deliverable (Ballard, 2000; Kerzner, 2022).

CHAPTER THREE

3. PPC systems, inventory/procurement, scheduling, and capacity

3.1 Inventory and procurement in construction: from "stock" to "site-ready flow"

In construction, inventory is less related to that of finished goods, which is more about the availability of materials, workface readiness and supply risk. Having the same ordering logic on all items causes two predictable failures: (i) overstocking bulk items which limit the site space and increase the damage/handling loss; and (ii) stockout of critical, long-lead items which block the critical path (Slack, Brandon-Jones and Johnston, 2019).

One viable consultant solution to NordBuild Construction GmbH (NBC) is dividing procurement by the ABC analysis according to the schedule criticality x cost x lead-time risk instead of price (Stevenson, 2021):

A-items (high criticality / long-lead / high risk): facade, elevators, switchgear, HVAC, structural steel. Policy: early buying and milestone-based supplier guarantees, official expediting causes, and contingencies (dual sourcing or approved practices).

B-items (medium): rebar, low standard MEP components, doors, windows (according to the project). Policy: framework contracts, scheduled deliveries consistent to lookahead schedules, moderate buffers.

C-Items (low-criticality/high-availability): fasteners, ordinary consumables, small tools. Policy: ordering and replenishment simplified to decrease the transaction cost (Stevenson, 2021).

JIT and Kanban may be applied in construction, but it requires adapting them to the reality of the site. The zero inventory ideology is dangerous as the suppliers and transportation are uncertain and the NBC ought to implement JIT at the work face level by delivery of materials just in time to the point of utilization which is backed up by stable lookahead plan, secure logistics windows and suppliers who are dependable (Koskela, 2000). In the case of repetitive materials (e.g. drywall boards, standard conduit), Kanban-type replenishment has the potential to reduce the

work of the administration and avoid stagnation of work, however, there must be a disciplined storage and an indication of a reorder (Slack, Brandon-Jones and Johnston, 2019). NBC safety stock logics must be discriminate and must be linked with uncertainty: buffers are reasonable when trading A-item and when the cost of waiting one day is higher than that of having an additional stock (Hopp and Spearman, 2011).

3.2 Planning systems: mapping MRP-I / MRP-II / ERP to construction

MRP-I reason (materials planning): break down the Bill of Quantities (BoQ), and work packages into time-phase material requirements based on the schedule (Vollmann et al., 2005).

MRP-II logic (materials + capacity): combine material plans and capacity availability (crews, cranes, subcontractors, site management) to check the feasibility and not to establish the procurement commitments on the work that cannot be performed (Vollmann et al., 2005).

ERP (enterprise integration): integrates estimating, procurement, finance, contract administration, project controls, and HR in such a way that the decisions and performance reporting are governed by a single set of numbers (Slack, Brandon-Jones and Johnston, 2019).

Consultant advice: NBC needs to adopt a planning cadence in which (i) a master resource plan is based on portfolio forecast; (ii) each project has a rolling lookahead plan; and (iii) procurement releases are not only based on the baseline schedule dates but also based on constraint readiness (design available, predecessor complete, access approved). This helps in eliminating premature procurement, site congestion as well as safeguarding the critical path (Ballard, 2000; Kerzner, 2022).

3.3 Scheduling: CPM/Gantt plus constraint-based short-horizon control

NBC requires two layers of scheduling model: Baseline schedule (contract + governance): 1.CPM. CPM resolves the critical path, float, and milestone risks; it is imperative in reporting to clients and claims management (Kerzner, 2022). Gantt charts transfer this into a plan that can be communicated and coordination of the trades and stake holders is achievable. 2.Constrained and lookahead scheduling (execution control): CPM is poor as a day to day prevailing production control in uncertainty conditions. NBC is recommended to have running lookahead schedules (e.g., 6-week windows) where tasks are released only when the constraint has been cleared - at the heart of the Last Planner System (Ballard, 2000). This addresses one of the major

construction failure modes i.e. crews are allocated tasks which are not yet ready leading to waiting, rework and make-do (Koskela, 2000). 3.Bottleneck prioritisation: Constraints can be located at bottlenecks in NBC in a scarce trades (MEP coordination), crane availability, inspections or design approvals. These constraints should be taken seriously by NBC in order to avoid unplanned queues and idleness- in line with the fact that system performance is determined by the constraint rather than average capacity (Hopp and Spearman, 2011).

3.4 Capacity: integrating RCCP and CRP equivalents for construction

RCCP equal (portfolio level, monthly / quarterly): Verification of checks and balances against these critical resources: key trade crew-months, crane count, site managers, major subcontractor capacity. Lock short term commitments and leave long horizons adaptable using time fences (Vollmann et al., 2005).

CRP equal (project level, every week): Check short-term feasibility of lookahead window: Does the lookahead window require crews, equipment and materials, are they available? Otherwise, the re-ordering of activities has to be done prior to implementation (Ballard, 2000).

Short term capacity plans: overtime, shift alteration, temporary labour, subcontractor surge capacity, resequencing, and selective buffering.

Long-term strategies: framework contracts with the most important subcontractors, talent pipeline management, prefabrication/offsite techniques to make on-site labour less intensive, and portfolio smoothing using start-date governance (Slack, Brandon-Jones and Johnston, 2019; Kerzner, 2022).

Integration (what markers want): inventory choices have impacts on site congestion and task preparedness; procurement lead times have impacts on the feasibility of schedules; capacity scarcity influences what can be committed at MPS/portfolio level.

CHAPTER FOUR

4.Digitalisation and IT systems in NBC

4.1 ERP and APS: turning PPC into a controlled, data-driven system

Digitalisation must not be put as technology equals performance. For NBC:

- ERP combines procurement, finance, HR, contract management, and project controls, which allows a uniform cost/schedule information, supplier performance monitoring, and project-wide governance (Slack, Brandon-Jones and Johnston, 2019). In the absence of ERP discipline, PPC is disjointed spreadsheets and incompatible numbers.

- The concepts of APS (Advanced Planning and Scheduling) can be applied in cases when the NBC should organize the allocation of scarce shared resources (cranes, specialist crews, prefabrication slots). APS can help in plan the scenario and levelling of resources, but only when the inputs are trusted and the active regulations are obeyed (Vollmann et al., 2005).

- Consultant position: Introduce ERP as the source of truth, and use APS selectively on resource constraints in portfolio resources, and not on the overly complex optimisation everywhere.

4.2 Industry 4.0 enablers: BIM, IoT, digital twins, and analytics (construction-fit)

In construction, the most potent digital lever is usually the BIM (Building Information Modeling) as it is interconnected with design, quantities, sequencing, and change management (Eastman et al., 2018). In PPC terms, BIM improves:

1. Quantity accuracy (enhanced dependent demand planning / BoQ conviction).

2. Clash detection (fewer reworks, fewer crazy schedules).

3. 4D planning (timeline-based visualisation of models that enhance stakeholder going-on).

4. Change control (propagation of changes throughout the quantities and plans).

Follow-up and control can be enhanced with IoT and site data capture (e.g., equipment utilisation, delivery tracking, environmental sensors) at the expense of subjective status reporting by giving near-real-time indications of progress (Slack, Brandon-Jones and Johnston, 2019). A practical method is to begin with high value use cases:

1.Monitoring the deliveries of A-items (facade units, switchgear) within schedule.

2.Tracking the use of cranes and time missed.

3.Recording productivity indicators of constraint trades.

The concept of digital twins must be presented cautiously: it can be valuable when organisations have predictable data streams, well-defined choices they can back (e.g. scenario testing can be used to test sequencing and resource conflicts). Otherwise, they may become costly to visualise and have little effect in terms of operations (Eastman et al., 2018).

4.3 Human and ethical dimensions: cybersecurity, data governance, and resistance to change

1. Cybersecurity: construction supply chains are characterized by a number of external companies (subcontractors, designers). The growing level of digitalization is creating bigger attack surfaces (phishing, ransomware, compromised accounts).

2. Data governance: a single number needs to be defined (rules of progress measurement, cost coding, change order classification). Bad PPC decisions and false confidence are the results of poor data governance (Slack, Brandon-Jones and Johnston, 2019).

3. Resistance to change and capacity building: the digital PPC tools transform the ways in which foremen, planners, and subcontractors work. NBC needs to invest into training and refocus incentives (e.g. plan reliability targets, accountability of constraint removal) to make people feel the system as enabling them and not policing them (Ballard, 2000).

4. Task 4: Digitalisation Consultant conclusion Digitalisation enhances PPC in cases where it enhances constraint visibility, plan reliability and disciplined decision-making.

CONCLUDING REMARKS

At NordBuild Construction GmbH (NBC), work is highly interconnected, demand is unpredictable, and limited resources are more important than ideal timelines. According to this research, PPC need to be a strategic competency that uses integrated portfolio planning, procurement, capacity management, and site control to transform market uncertainty into dependable commitments (Slack et al., 2019; Kerzner, 2022). For NBC, a two-tier PPC strategy works effectively. Probability-weighted forecasts and time fences on a portfolio level stabilise the commitments and ensure overbooking is avoided, whereas RCCP-style checks expose the bottlenecks at an early stage to allow the subcontractor structures, start-date governance, and prefabrication options to be activated (Stevenson, 2021; Vollmann et al., 2005). At project level, CPM is still needed to ensure contractual control, and predictable delivery is necessary to ensure such protection of workflow and minimize making-do, waiting and rework (Ballard, 2000; Koskela, 2000). To ensure that buffers protect important channels without clogging sites, inventory should be divided into segments based on lead time and criticality (Hopp and Spearman, 2011). According to Eastman et al. (2018), digital technologies like as ERP and BIM are only beneficial when they enhance decision quality and execution dependability through appropriate governance and user engagement, not as technology in and of itself.

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APPENDIX (if necessary)